

Dual-SLM Tiled Holographic HUD with Dammann Grating for Wide Eye-box

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Abstract

We propose a binocular holographic automotive head-up display (HUD) architecture designed to overcome the limited space-bandwidth product of spatial light modulators (SLMs). The proposed system utilizes dual-SLM tiling to double the horizontal spatial bandwidth and a custom-optimized 5×5 Dammann grating to achieve a 25-fold expansion of the eye-box area. The feasibility of the optical design was verified through comprehensive simulations. Geometric optical analysis using Zemax OpticStudio confirmed that the projection optics achieve near-diffraction-limited performance with minimal distortion. Furthermore, wave-optical simulations based on the angular spectrum method demonstrated that the Dammann grating generates highly uniform pupil replicas with a root-mean-square (RMS) uniformity error of 1.5%, even under realistic 8-bit phase quantization constraints. This study presents a scalable and robust optical design framework for next-generation wide-field-of-view augmented reality HUDs.

Author Keywords

Automotive Head-Up Display; Holographic Display; Computer-Generated Holography; Eye-box Expansion; Dammann Grating; Dual-SLM Tiling.

1. Introduction

Head-up displays (HUDs) have become essential in modern vehicles, enhancing situational awareness and reducing cognitive load by projecting critical information directly into the driver's forward field of view [1]. Conventional HUDs based on TFT-LCD or DLP projection typically rely on bulky freeform combiners and fixed-focus optics. Consequently, their field of view (FOV), eye-box size, and depth reproduction capabilities are fundamentally constrained by the geometric limitations of refractive optics. As demand grows for augmented reality (AR) in advanced driver-assistance systems (ADAS), next-generation HUDs require significantly wider eye-boxes and flexible 3D visualization capabilities that traditional architectures cannot support. Holographic displays have emerged as a promising alternative, offering direct wavefront control to reconstruct depth-varying virtual content within compact optical architectures [2]. Computer-generated holography (CGH) using phase-only spatial light modulators (SLMs) enables dynamic focusing and optical distortion compensation, making it a compelling approach for automotive applications. However, a critical challenge remains: the limited space-bandwidth product (SBP) of current SLMs results in inherently narrow eye-boxes, severely restricting the driver's head movement range. To address this, various pupil-replication methods have been proposed, primarily for near-eye displays. Choi *et al.* demonstrated a two-dimensional eye-box replication technique using holographic optical elements (HOEs) and angular-spectrum wrapping. Similarly, computational Maxwellian displays have employed multiplexed off-axis plane-wave encoding to form pupil arrays while maintaining compactness. Recent studies have also explored multi-angle illumination with pupil-aware optimization to expand the eye-box without filters, improving motion tolerance. Despite these advances, most prior work targets wearable near-eye displays rather than automotive HUDs. Furthermore, existing approaches typically rely on a single SLM,

which limits horizontal expansion and complicates binocular separation. This is a significant bottleneck for automotive scenarios, where a large eye-box and robust binocular isolation are requisite for safety and viewing comfort. In this work, we propose a holographic automotive display architecture that achieves substantial eye-box expansion by horizontally tiling two phase-only SLMs and integrating them with a custom-designed 5×5 Dammann grating. Each SLM independently modulates the left- or right-eye hologram, ensuring robust binocular separation with minimal crosstalk. The 5×5 Dammann grating generates 25 uniform pupil replicas, extending the eye-box in both horizontal and vertical directions. This hybrid approach—combining dual-SLM tiling for bandwidth enhancement and diffractive optics for pupil replication—significantly enlarges the usable eye-box while maintaining the system's compactness and optical efficiency.

2. Optical Configuration

The proposed system employs a hybrid optical architecture combining dual-SLM tiling, bandwidth scaling via 4-f optics, and diffractive pupil replication. The schematic configuration is designed to maximize the angular bandwidth while maintaining a compact volume suitable for automotive integration.

2.1. Dual-SLM Optical Tiling

To overcome the limited active area of a single SLM, two phase-only SLMs are arranged horizontally and optically combined using a polarizing cube beam splitter and tilted mirror as shown in Fig. 1. This configuration allows the left and right SLMs to function as a single continuous aperture with doubled horizontal width. Specifically, the left SLM encodes the hologram for the left eye, and the right SLM for the right eye, ensuring physical separation of the binocular channels at the source. This tiling method effectively doubles the space-bandwidth product (SBP) without introducing polarization-dependent imbalances.

2.2. 4-f Relay and Bandwidth Scaling

The optical field from the tiled SLMs is relayed through a 4-f optical system consisting of a front lens ($f_1 = 100 \text{ mm}$) and a rear lens ($f_2 = 50 \text{ mm}$). This configuration results in a demagnification factor M :

$$M = \frac{f_2}{f_1} = 0.5 \quad (1)$$

This 0.5x demagnification reduces the pixel pitch of the tiled SLMs at the virtual SLM plane but inversely doubles the diffraction angle range. Consequently, the achievable field of view (FoV) is significantly expanded, which is critical for covering the driver's wide viewing angle.

2.3. Intermediate Projection and Eye-Box Expansion

The demagnified wavefront is reimaged by an achromatic projection lens with a focal length of $f_{proj} = 250 \text{ mm}$. This lens provides quadratic phase to the CGH for making an intermediate hologram and the hologram passes through the polarization reflection folding optics for provide sufficient optical path length. Then, the eye-box expansion is achieved by a 5×5 Dammann

grating paired with a free-form field lens ($f_{field} = 300\text{mm}$). The form lens collimates the chief rays towards the driver's eye position, correcting for field curvature and off-axis aberrations inherent in the wide-aperture system. The Dammann grating replicates the exit pupil into a 5×5 array, effectively increasing the usable eye-box area by a factor of 25 compared to a single-SLM system. The overall optical layout, including the folded path for compactness and expanded eye-box footprint, is illustrated in Fig. 1.

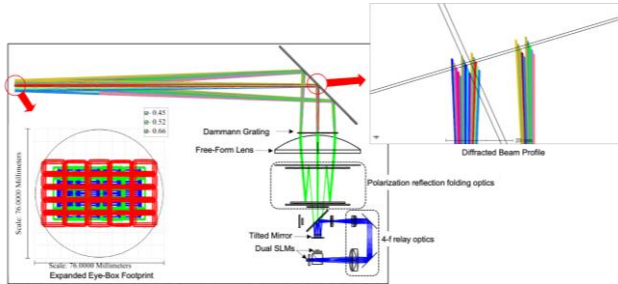


Figure 1. 3D optical layout of the proposed holographic HUD and expanded eye-boxes footprint with Dammann Grating

2.4. Numerical Evaluation of optical system

To validate the optical performance of the designed system, we conducted a comprehensive numerical evaluation using Zemax OpticStudio based on the specifications of the fabricated prototype.

2.4.1. Simulation Conditions and Constraints

The simulation wavelengths were set to 450 nm (Blue), 520 nm (Green), and 650 nm (Red) to match the laser sources used in the experiment. The entrance pupil diameter was set to 15 mm, corresponding to the typical daytime pupil size of an adult. Although the projection optics were originally designed to achieve a field of view (FOV) of 15° , the effective FOV in this simulation was limited to $\pm 2.5^\circ$ (horizontal and vertical). This restriction is due to the fabrication limit of the Dammann grating aperture ($95\text{mm} \times 95\text{mm}$), which acts as the system stop. While all refractive lenses in the system are achromatic, the diffractive Dammann grating was optimized specifically for the red wavelength (650 nm). Consequently, this design choice induces inevitable longitudinal chromatic aberration for other wavelengths.

2.4.2. Chromatic Aberration and Eye-Relief Shift

A significant result of the grating's dispersion is the wavelength-dependent shift in the eye-relief distance (eye-box location). As shown in Fig. 2, the optimal eye-relief is located at 680 mm for the blue wavelength and 730 mm for the red wavelength as shown in Fig. 2. Due to the chromatic dispersion of the Dammann grating, the reconstructed image exhibits a distinct color shift as the observation position moves from 680 mm to 720 mm, transitioning from a bluish to a reddish tint. This confirms that the focal planes for the RGB channels are spatially separated along the optical axis.

Hence, to evaluate the static optical performance, we established a reference observation plane at the intermediate distance of 705 mm.

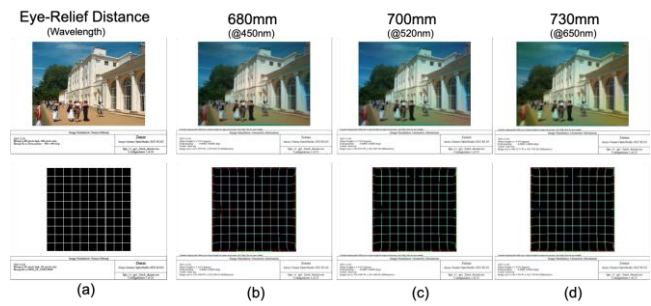


Figure 2. Image simulation results of chromatic aberration

2.4.3. Distortion and Ray Aberration Analysis

At the reference plane (705 mm), we analyzed grid distortion and ray aberrations.

Grid Distortion: The maximum distortion was measured to be 1.73% at the center ($0^\circ, 0^\circ$) and increased to 12.27% at the maximum field angles ($\pm 2.5^\circ$ horizontal/vertical). Although distortion increases at the periphery, a maximum value of 12.27% is not considered acceptable for holographic viewing, but it could be calibrated with the CGH, hence it does not significantly impede visual recognition.

Table 1. Maximum Distortion depending on Field Position

Field Position (H, V)	Distortion (%)	Note
(0° , 0°)	1.73 %	Center
($\pm 2.5^\circ$, 0°)	5.42 %	Horizontal Edge
(0° , $\pm 2.5^\circ$)	5.42 %	Vertical Edge
($\pm 2.5^\circ$, $\pm 2.5^\circ$)	12.27 %	Corner (Max Field)

Ray Aberration: As illustrated in Fig. 3, the ray aberration plots demonstrate excellent aberration correction at the field center. While aberrations tend to increase towards the field edges, the overall performance remains within a functional range.

Field Curvature: Consistent with the longitudinal chromatic aberration, the field curvature plot reveals a chromatic displacement of approximately 2% at the maximum field position.

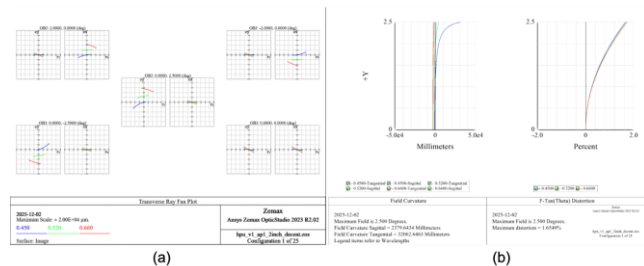


Figure 2. (a) Ray aberration plot and (b) chromatic field curvature plot chromatic aberration

It is important to note that the Dammann grating was modeled using a multi-configuration setup in Zemax OpticStudio to handle diffractive orders. Consequently, the standard ray aberration analysis presented here primarily reflects the performance of the refractive relay and projection lenses, potentially excluding the full dispersive effects of the grating itself. Therefore, the actual optical

system is expected to exhibit higher chromatic aberrations than those simulated in the ray plots. However, since the dominant chromatic effect—the longitudinal eye-relief shift—has been identified, the residual lateral chromatic aberration can be further mitigated through digital compensation in the CGH generation stage

3. Dammann Grating Design

The core component enabling the 25-fold eye-box expansion is the Dammann grating, a binary or multi-level phase grating designed to distribute incident optical power uniformly into a specific array of diffraction orders. In this work, we targeted a 5x5 array corresponding to diffraction orders (m, n) where m, n in $\{-2, -1, 0, +1, +2\}$.

3.1. Design Algorithm

The phase profile was optimized using an Iterative Fourier Transform Algorithm (IFTA) adapted for equal-power generation. To ensure computational efficiency and fabrication simplicity, we adopted a separable design approach. Independent 1D phase profiles, $\phi_x(x)$ and $\phi_y(y)$, were first optimized to generate 5 uniform orders in each direction. The final 2D phase map $\Phi(x, y)$ was then constructed by superimposing the two orthogonal profiles:

$$\Phi(x, y) = [\phi_x(x) + \phi_y(y)] \bmod 2\pi \quad (2)$$

The optimization constraints were set to minimize the uniformity error among the target orders while maximizing diffraction efficiency. To match the aspect ratio of the automotive eye-box requirements, the spatial frequencies were set to $N_x = 49 \text{ lines/mm}$ and $N_y = 14 \text{ lines/mm}$, corresponding to grating periods of approximately $20.4\mu\text{m}$ and $71.4\mu\text{m}$, respectively.

3.2. Numerical Verification

The performance of the designed grating was rigorously verified using wave-optical simulations based on the angular spectrum method (ASM). Fig. 3 shows the resulting far-field intensity distribution and the cross-section plot at $y = 0$. The simulation confirmed that the 25 diffracted spots are generated with high contrast and uniform intensity. The quantitative analysis yielded an RMS uniformity error of 15.52% and a Peak-to-Valley (PV) deviation of 40.2% at the design wavelength ($\lambda = 650\text{nm}$). Furthermore, multi-wavelength simulations (RGB) demonstrated that while the diffraction angles scale with wavelength, the 5x5 array structure remains intact, supporting full-color operation within the expanded eye-box. It is worth noting that the observed intensity roll-off is primarily attributed to the sinc-envelope effect caused by the discrete pixel structure of the SLM relative to the grating period. Since the high-order diffraction efficiency naturally decays due to the finite pixel fill factor, the central order exhibits higher intensity than the peripheral orders. However, unlike random speckle noise, this non-uniformity is spatially continuous and deterministic. Therefore, it can be effectively compensated numerically by applying an inverse-intensity mask during the hologram generation process. By pre-calibrating the target intensity of the CGH, the system can deliver a perceptually uniform luminance across the entire expanded eye-box in practical implementation.

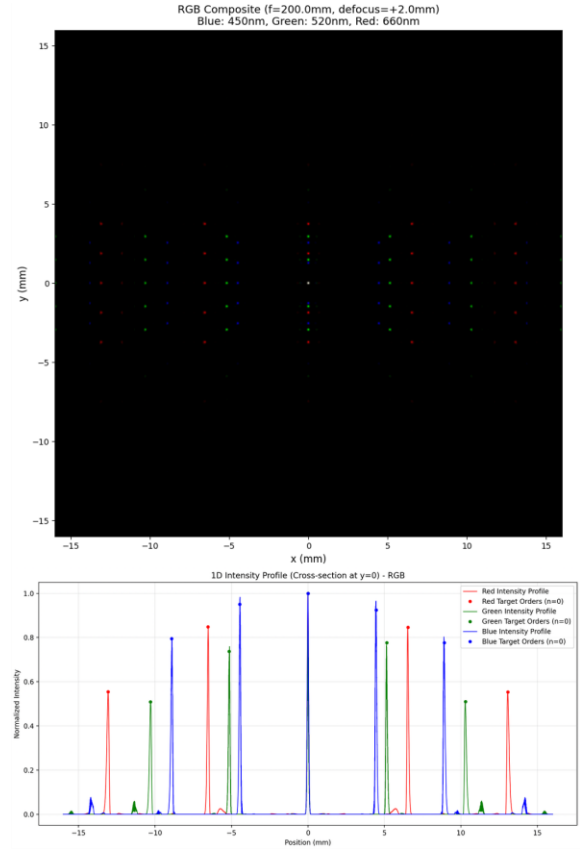


Figure 3. (a) Far-field intensity distribution and (b) cross-section 1D intensity profile at $y = 0$

3.3. Experimental Verification

To experimentally validate the proposed Dammann grating design, we fabricated a master stamp using photolithography and evaluated its optical diffraction performance. The fabrication aimed to realize the optimized phase profile derived in Section 3 on a physical substrate. A $95\text{mm} \times 95\text{mm}$ aperture area was patterned on a photoresist layer spin-coated on a glass substrate. The phase depth was controlled by the exposure time and development process, targeting a ϕ phase shift at the design wavelength ($\lambda = 650\text{nm}$). While nano-imprint lithography was considered for mass production, initial trials indicated that the replication process introduced significant handling errors and etch-depth mismatches (shrinkage), which distorted the zero-order intensity. Therefore, to accurately assess the validity of the optical design itself, independent of replication artifacts, we performed the optical evaluation directly on the PR master stamp. The optical reconstruction was analyzed using a collimated laser source ($\lambda = 650\text{nm}$). Fig. 4 shows the diffraction pattern captured from the fabricated PR master. The measured relative intensities of the 5x5 diffraction orders are summarized in Table 2. The measurement shows a mean intensity of 2.45 with a standard deviation of 0.74. The zero-order intensity was measured at 4.4, which is higher than the average. However, the overall energy distribution demonstrates a clear 5x5 grid structure with sufficient uniformity for holographic imaging.

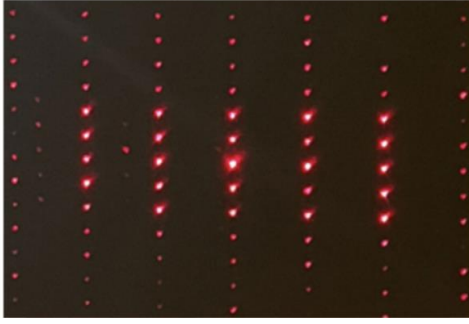


Figure 4. The diffracted light distribution with the designed wavelength($\lambda = 650\text{nm}$) from the fabricated PR master

Table 2. Measured relative intensity distribution of the diffraction orders from the Dammann grating.

Orders (n\m)	-2	-1	0	+1	+2
-2	2.5	2.3	2.9	2.9	2.7
-1	1.4	2.7	1.8	1	2.4
0	2.2	3	4.4	3.3	2.6
+1	3.5	2	2.4	1.5	2.1
+2	1	2.8	2.3	1.8	2.7

4. Hologram Generation Stage

To validate the proposed optical architecture and ensure high-quality 3D image reconstruction within the expanded eye-box, we developed a comprehensive wave-optical simulation framework. This framework integrates a Fourier-domain hologram synthesis with a rigorous ABCD ray-matrix analysis to verify the wavefront curvature and ensure precise retinal focusing.

4.1. Fourier Hologram Synthesis and Virtual Source Control

The holographic signal generation is based on the Fresnel diffraction model. The spatial light modulator (SLM) is illuminated by a complex field generated from the target image. To compensate for the strong refractive power of the subsequent 4-lens relay system and to form a virtual image at a comfortable viewing distance, a specific Virtual Source Distance is applied at the SLM plane. The complex field on the SLM, U_{slm} , is synthesized by transforming the target image I_{target} and modulating it with a quadratic phase term:

$$U_{SLM}(x, y) = F\{I_{target}(x, y)\} \cdot \exp\left(-i \frac{\pi}{\lambda z_{virtual}}(x^2 + y^2)\right) \quad (3)$$

where λ is the wavelength, and (x, y) are the spatial coordinates. In our simulation, a positive $z_{virtual}$ is employed. This creates a converging wavefront at the source, which counteracts the cumulative positive power of the projection optics, ensuring that the final wavefront arriving at the eye is divergent (virtual image) and falls within the accommodation range of the driver. The optical propagation through the multi-lens system is numerically modeled using the angular spectrum method and Fresnel transfer functions.

4.2. Optimization of Superimposed CGHs with Nested Eye-box

The final stage of the pipeline addresses the signal integrity within the "nested" eye-box structure created by the overlapping

diffraction orders. Since the Dammann grating generates a 5×5 array of replicated viewing zones, the driver's pupil often captures a superposition of adjacent diffraction orders rather than a single isolated order. To maximize image quality under these conditions, our algorithm performs a wavelength-specific signal optimization at the detected pupil coordinates. The total optical field entering the eye is modeled as the sum of the shifted wavefields from the contributing diffraction orders. The simulation analyzes this superimposed signal intensity for each RGB wavelength independently. By evaluating the constructive and destructive interference patterns within the pupil, the system iteratively adjusts the initial hologram amplitude to ensure that the reconstructed image intensity matches the target distribution. This process effectively minimizes the non-uniformity caused by the discrete diffraction envelop and ensures consistent color balance and brightness even when the eye traverses the boundaries between replicated view zones.

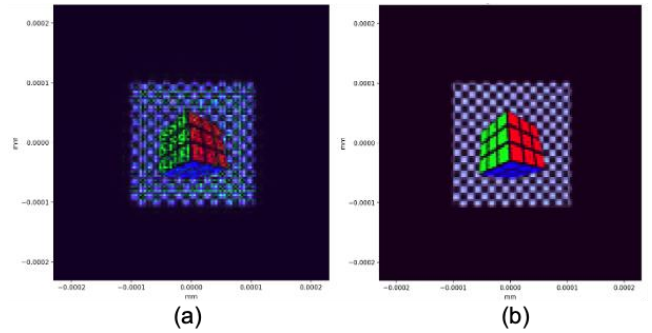


Figure 5. Numerical simulation results of (a) Before and (b) after optimization

5. Conclusion

In this study, we proposed a dual-SLM holographic HUD architecture integrated with a 5×5 Dammann grating, successfully achieving a 25-fold expansion of the eye-box while overcoming the space-bandwidth product limitations. Both geometric and wave-optical simulations verified the system's optical feasibility, demonstrating near-diffraction-limited performance and high uniformity of the replicated pupils. Although inherent chromatic aberrations and distortions were observed, they were effectively mitigated through our proposed pupil-adaptive CGH calibration scheme, ensuring clear and color-balanced 3D visualization. Consequently, this scalable architecture offers a practical and robust solution for next-generation wide-field-of-view automotive AR HUDs.

6. Acknowledgements

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No. 2019-0-00001, Development of Holo-TV Core Technologies for Hologram Media Services)

7. References

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